

Biophysical animal models

Day 2: From biophysics to macroecology

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Biophysical ecology applies the principles of thermodynamics to organisms to derive mechanistic models of energy and mass exchange (Gates, 1980; W. P. Porter et al., 1973; W. P. Porter & Gates, 1969).

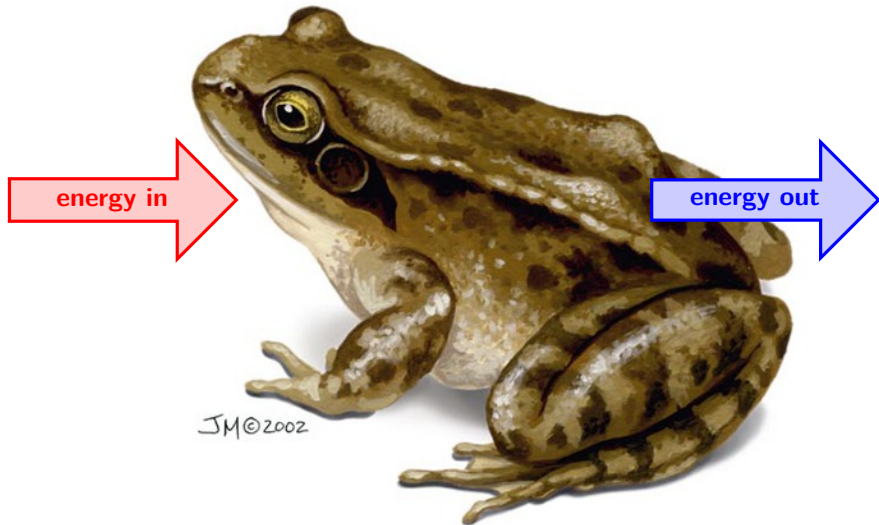
Core idea

Organisms are open thermodynamic systems exchanging heat, water, and gases with their environment. These exchanges can be described by the conservation laws of physics.

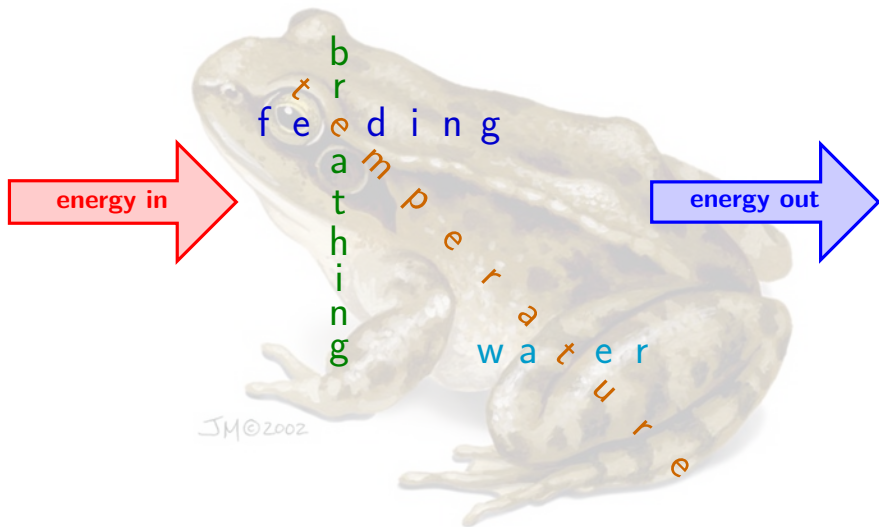
The four domains of energy and mass balance



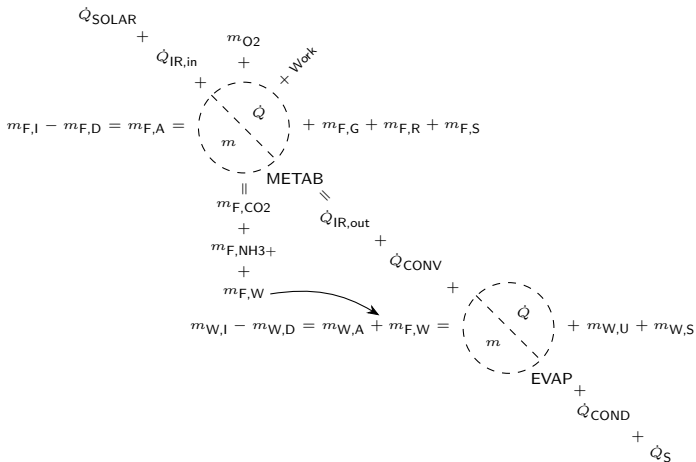
The four domains of energy and mass balance



The four domains of energy and mass balance



Coupled energy and mass balance



Coupled energy and mass balance

Organisms must simultaneously satisfy energy and mass balance:

$$\dot{Q}_{solar} + \dot{Q}_{IR,in} + \dot{Q}_{metab} = \dot{Q}_{IR,out} + \dot{Q}_{conv} + \dot{Q}_{evap} + \dot{Q}_{cond} + \dot{Q}_s$$

Energy balance determines:

- body temperature (ectotherms)
- metabolic rate (endotherms)
- activity and performance

Mass balance determines:

- water status
- energy reserves
- growth and reproduction

Based on M. Kearney and Porter (2009)

The fundamental niche from first principles

- An organism cannot survive where it would be in negative energy or mass balance
- Reproductive capacity depends on “profits” in these currencies
- The set of environmental conditions where energy and mass balance is positive defines the **fundamental niche**

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Key insight

The fundamental niche can be **computed** from first principles — it is not merely a statistical description of where a species is found.

From functional traits to climate space

An organism's functional traits (morphology, physiology, behaviour) determine how it experiences a given set of environmental conditions:

Functional traits:

- body size and shape
- solar reflectivity (albedo)
- skin permeability
- metabolic rate
- behavioural repertoire

Environmental axes:

- air temperature
- solar radiation
- wind speed
- humidity
- substrate temperature

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The energy balance equation translates environmental conditions into body temperature — the key physiological variable for ectotherms.

Workflow for mechanistic niche modelling

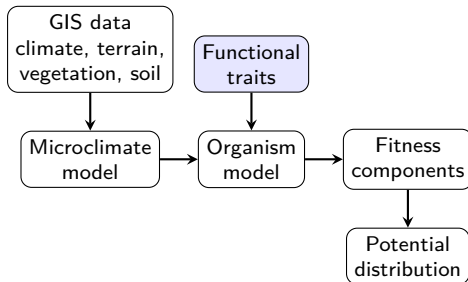


Figure 1: Workflow: GIS data → microclimate → organism model → fitness → distribution. Based on M. Kearney and Porter (2009).

Biophysical Ecology processes

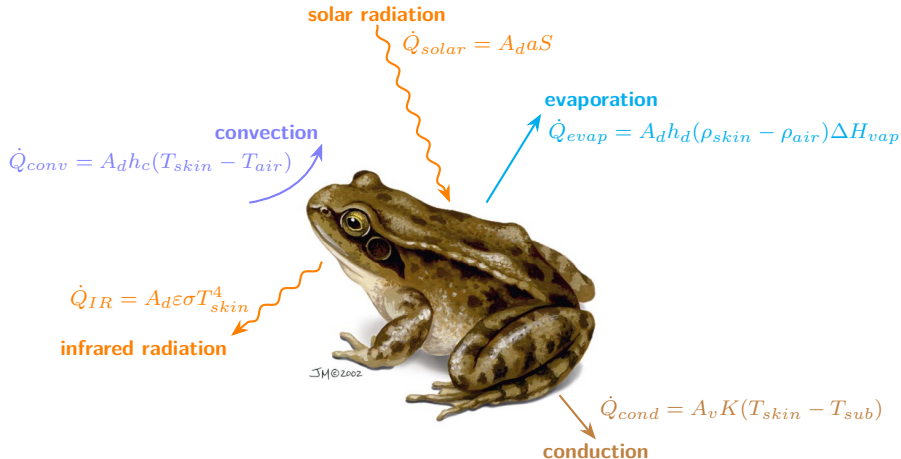


Figure 2: Heat transfer mechanisms for a frog.

Convective heat transfer

Convection is the transfer of heat between a surface and a moving fluid (air or water). In biophysical ecology, convection determines how quickly an organism exchanges heat with the surrounding air.

$$\dot{Q}_{conv} = A_d h_c (T_{skin} - T_{air})$$

Where:

- $\dot{Q}_{conv}$ = convective heat flux (W)
- A_d = characteristic surface area exchanging heat (m²)
- h_c = convective heat transfer coefficient (W m⁻² K⁻¹)
- T_{skin} = skin or body surface temperature (°C or K)
- T_{air} = air temperature (°C or K)

Key concepts

- Heat flows from warmer to colder regions.
- Wind speed strongly affects h_c :
 - low wind $\rightarrow$ weak convection
 - high wind $\rightarrow$ strong convection
- Small organisms often experience stronger convective effects because of their high surface-area-to-volume ratio.

Convection is critical for:

- thermoregulation
- overheating risk
- cooling efficiency
- boundary layer dynamics

For example, a lizard exposed to strong wind may cool rapidly even under direct sunlight.

NicheMapR implementation: convection

The effective convection area in `CONV_ecto` is $CONVAR = A_{tot} - A_v - A_t$ (total area minus ventral and touching surfaces). Free and forced convection are computed separately via Nusselt–Reynolds correlations, then combined into a single $\dot{Q}_{conv}$.

See `CONV_ecto.R` in the NicheMapR source.

Conductive heat transfer

Conduction is the direct transfer of heat between objects in physical contact.

$$\dot{Q}_{cond} = A_v K (T_{skin} - T_{sub})$$

Where:

- $\dot{Q}_{cond}$ = conductive heat flux (W)
- A_v = contact area with the substrate (m^2)
- K = thermal conductance coefficient ($\text{W m}^{-2} \text{K}^{-1}$)
- T_{skin} = body surface temperature ($^{\circ}\text{C}$ or K)
- T_{sub} = substrate temperature ($^{\circ}\text{C}$ or K)

- Heat transfer occurs through molecular collisions.
- The efficiency of conduction depends on:
 - thermal conductivity of the substrate
 - contact area
 - temperature difference
- Wet soils generally conduct heat better than dry soils.

Conduction is especially important for:

- burrowing organisms
- eggs in nests
- ectotherms resting on rocks or soil

For example, reptile eggs exchange heat conductively with the surrounding soil.

NicheMapR implementation: conduction

The conductance coefficient in `COND_ecto` is computed as K_{sub}/d where K_{sub} is the substrate thermal conductivity and d is the depth over which conduction occurs (default 2.5 cm).

See `COND_ecto.R` in the NicheMapR source.

Thermal infrared radiation

All objects with a temperature above absolute zero emit thermal infrared radiation.

$$\dot{Q}_{IR} = A_d \varepsilon \sigma T_{skin}^4$$

Where:

- $\dot{Q}_{IR}$ = emitted infrared radiation (W)
- A_d = radiative surface area (m²)
- ε = emissivity of the surface (dimensionless)
- σ = Stefan–Boltzmann constant

$$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$$

- T_{skin} = absolute skin temperature (K)

- Infrared radiation increases rapidly with temperature because of the fourth-power relationship.
- Organisms both:
 - emit infrared radiation
 - absorb infrared radiation from the environment
- Net infrared exchange depends on the temperatures of surrounding surfaces and the sky.

Infrared radiation strongly influences:

- nocturnal cooling
- desert thermoregulation
- nighttime frost exposure

Clear skies often increase radiative heat loss at night.

NicheMapR implementation: infrared radiation

Incoming IR is computed separately for sky and ground using configuration factors F_{atosk} (to sky) and F_{atosb} (to ground). Outgoing IR also uses F_{atosk} for the fraction of emitted radiation reaching the sky.

See `RADIN_ecto.R` and `RADOUT_ecto.R` in the NicheMapR source.

Absorption of solar radiation

Solar radiation is one of the main heat inputs for organisms exposed to sunlight.

$$\dot{Q}_{solar} = A_d a S$$

Where:

- $\dot{Q}_{solar}$ = absorbed solar heat flux (W)
- A_d = projected area exposed to radiation (m²)
- a = absorptivity of the surface (dimensionless)
- S = incoming solar radiation (W m⁻²)

- Solar radiation includes:
 - direct radiation
 - diffuse radiation
 - reflected radiation
- Dark surfaces usually absorb more solar energy than light surfaces.
- Body posture and orientation affect the projected area.

Solar radiation is fundamental for:

- basking behavior
- warming of ectotherms
- nest heating
- microhabitat selection

For example, reptiles regulate body temperature behaviorally by changing sun exposure.

Absorbed solar is the sum of:

- direct beam (via ASIL, ZEN)
- diffuse sky (via FATOSK)
- reflected from substrate (via ABSSB)

— each modulated by appropriate geometry factors.

See `SOLAR_ecto.R` in the NicheMapR source.

Evaporative heat loss

Evaporation removes heat because water requires energy to change from liquid to vapor.

$$\dot{Q}_{evap} = A_d h_d (\rho_{skin} - \rho_{air}) \Delta H_{vap}$$

Where:

- $\dot{Q}_{evap}$ = evaporative heat loss (W)
- A_d = evaporative surface area (m^2)
- h_d = mass transfer coefficient (m s^{-1})
- ρ_{skin} = water vapor density at the skin (kg m^{-3})
- ρ_{air} = water vapor density of the air (kg m^{-3})
- ΔH_{vap} = latent heat of vaporization (J kg^{-1})

Key concepts

- Evaporation always cools the surface.
- Dry air increases evaporation rates.
- Wind enhances evaporation by removing humid boundary layers.

Evaporative cooling is important for:

- amphibians
- sweating mammals
- panting birds
- plant transpiration

Water balance and heat balance are therefore tightly linked.

NicheMapR implementation: evaporation

SEVAP_ecto computes cutaneous, ocular, and oral evaporation separately. The effective skin area is modulated by `pct_wet`. Respiratory water loss is handled by `RESP_ecto`.

See `SEVAP_ecto.R` in the NicheMapR source.

Metabolic heat production

Metabolic processes generate heat as a by-product:

$$\dot{Q}_{metab} = f(M, T_b, \text{activity})$$

Where:

- M = body mass (kg)
- T_b = body temperature
- **Ectotherms:** usually negligible in the energy balance ($< 5\%$ of environmental fluxes)
- **Endotherms:** the dominant heat source, maintaining stable body temperature above ambient
- Scales with body mass and follows a Boltzmann–Arrhenius relationship with temperature

NicheMapR implementation: metabolism

MET_ecto uses the allometric-Arrhenius form:

$$\text{VO}_2 = M_1 M^{M_2} 10^{M_3 T} 10^{M_4}$$

(ml O₂ h⁻¹). Metabolic heat production is then:

$$\dot{Q}_{metab} = \frac{20.1}{3600} \times \text{VO}_2$$

plus an activity offset $\dot{Q}_{act}$ for active behaviours.

See MET_ecto.R in the NicheMapR source.

Organism classes in biophysical models

Biophysical models must be tailored to the organism's physiology:

Ectotherms (dry-skinned)

- steady-state energy balance
- no significant cutaneous water loss
- e.g. lizards, insects, snakes

Ectotherms (wet-skinned)

- coupled heat–water balance
- significant evaporative cooling
- e.g. amphibians, molluscs

Large organisms

- transient (time-dependent) balance
- thermal inertia buffers fluctuations
- e.g. large reptiles, mammals

Endotherms

- metabolic heat as energy source
- thermoneutral zone concept
- e.g. birds, mammals

Dry-skinned ectotherms

For reptiles, insects and many invertebrates, cutaneous water loss is negligible for the energy balance:

$$\dot{Q}_{solar} + \dot{Q}_{IR,in} = \dot{Q}_{IR,out} + \dot{Q}_{conv} + \dot{Q}_{cond}$$

- Solved as a **steady-state** balance (operative environmental temperature)
- Core temperature stabilises quickly due to low thermal inertia
- Key traits: solar reflectivity, surface areas, convection coefficients
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Classic approach

Empirical physical models (copper casts, taxidermic mounts) capture the thermal properties and are placed in the field to measure operative temperature.

Wet-skinned ectotherms

Amphibians and other wet-skinned organisms have significant evaporative heat loss:

$$\dot{Q}_{solar} + \dot{Q}_{IR,in} = \dot{Q}_{IR,out} + \dot{Q}_{conv} + \dot{Q}_{cond} + \dot{Q}_{evap}$$

- Coupled **heat and water balance**
- Evaporative cooling can keep body temperature *below* air temperature
- Key trait: fraction of skin acting as a free-water exchanger
- Water balance often more limiting than thermal balance

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Methods for partitioning cutaneous and respiratory water loss are detailed by Tracy (1976) and generalised in M. Kearney and Porter (2004).

Large organisms and transient effects

As body size increases, thermal inertia becomes important:

$$\frac{dT_b}{dt} = \frac{\dot{Q}_{net}}{c_p M}$$

Where:

- T_b = body temperature ($^{\circ}\text{C}$ or K)
- c_p = specific heat capacity of tissue ($\text{J kg}^{-1} \text{K}^{-1}$)
- M = body mass (kg)
- Temperature changes more slowly in larger animals (lower surface-area-to-volume ratio)
- Environmental conditions may change faster than body temperature
- Requires solving the **transient** (time-dependent) energy balance
- Relevant for large reptiles, endotherms, and fine temporal scales

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Solutions have been proposed by Seebacher and Alford (1999), Spotila et al. (1973), and Stevenson et al. (1985).

Endotherms

Birds and mammals produce sufficient metabolic heat to maintain a stable, elevated body temperature:

- Solve for the **metabolic rate** required to maintain homeothermy under given conditions
- Below the thermoneutral zone, metabolic rate increases to compensate for heat loss
- Above it, evaporative cooling (panting, sweating) is activated
- Insulation (fur, feathers, fat) modifies heat exchange

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Key outputs:

- Minimum and maximum metabolisable energy requirements
- Water requirements for evaporative cooling
- Critical environmental limits for survival

Aquatic organisms

For organisms permanently under water, heat exchange is dominated by convection:

- Body temperature closely tracks water temperature (exceptions: some endothermic fish and marine mammals)
- Momentum balance (swimming costs) can dominate energy budgets
- Locomotion in water is much more costly than in air
- Water availability is not limiting (unlike terrestrial organisms)

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Intertidal organisms face a unique challenge:

- Periodic shifts between aerial and aquatic conditions
- Wave splash effects on heat and mass exchange
- Detailed model: e.g., Helmuth (1998)

Different questions, different models

Computing body temperature from environmental conditions requires solving the energy balance equation, but the right approach depends on the question:

Steady-state: $\sum Q = 0$ each time step

- Instantaneous thermal equilibrium
- Appropriate for small ectotherms
- Behaviour as deterministic rules

Transient: $dT_b/dt = f(\dot{Q}_{net})$

- Thermal inertia buffers fluctuations
- Necessary for large organisms
- Behaviour as probabilistic choice

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Here we compare two implementations: NicheMapR's steady-state model and Rubalcaba et al.'s transient approach.

Body size gradients in ectotherms

Question: Do biophysical mechanisms explain latitudinal body size gradients in ectotherms (e.g. Bergmann's rule)?

Bergmann's rule (endotherms):

- Larger body size at higher latitudes
- Better heat retention (surface-to-volume)

Ectotherms:

- Complex, often contradictory patterns
- No single mechanism established
- Heat balance *and* water balance may interact

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Rubalcaba et al. (2019) built a biophysical model to derive null predictions of how body size should vary with latitude based purely on heat and water balance Rubalcaba et al. (2019).

The transient heat balance equation

Rubalcaba et al. used a **dynamic** (time-dependent) energy balance:

$$\frac{dT_b}{dt} = \frac{\dot{Q}_{sol} + \dot{Q}_{IR} + \dot{Q}_{conv} + \dot{Q}_{cond} + \dot{Q}_{evap}}{c_p M}$$

Where:

- $\dot{Q}_{sol}$ = absorbed solar radiation (W)
- $\dot{Q}_{IR}$ = net infrared radiation (W)
- $\dot{Q}_{conv}$ = convective heat flux (W)
- $\dot{Q}_{cond}$ = conductive heat flux (W)
- $\dot{Q}_{evap}$ = evaporative heat loss (W)
- c_p = specific heat capacity ($\text{J kg}^{-1} \text{K}^{-1}$)
- M = body mass (kg)

Implications of thermal inertia

- Body mass M and specific heat c_p determine thermal inertia
- Rate of temperature change $\propto$ surface-to-mass ratio
- Larger animals warm up and cool down *more slowly*

Key consequence

Large ectotherms attain higher body temperatures when basking and retain heat longer — an advantage in cold climates but a risk of overheating in warm climates.

Behavioural model: probabilistic choice

Animals select among microenvironments and postures to optimise thermal performance:

$$p(w_i) = \frac{\exp(\lambda w_i)}{\sum_{j=1}^n \exp(\lambda w_j)}$$

Repertoire (6 behaviours):

- 3 shade levels (0%, 50%, 90%)
 - 2 postures (100% or 40% exposed area)
- maximise $w(T)$ = thermal performance

λ parameter:

- $\lambda = 0$: random (non-thermoregulatory)
- $\lambda > 0$: thermoregulatory optimisation
- high $\lambda \rightarrow$ effective thermoregulation

Rubalcaba et al. (2019)

Key findings

Heat balance:

- Large ectotherms achieve higher T_b at temperate latitudes
- But overheat risk at southern latitudes
- Body size advantage vanishes at boreal latitudes
- Predicts Bergmann-like clines in temperate regions

Water balance:

- EWL decreases nonlinearly with body size
- Larger wet-skinned ectotherms conserve water better
- Predicts larger body size in dry southern areas
- Hydoregulation may drive converse Bergmann patterns

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Implication

Biophysical null models can generate testable predictions about macroecological patterns without requiring species occurrence data.

The NicheMapR ectotherm model

The model solves the coupled heat and water balance for an ectotherm at each time step, given microclimate inputs (M. R. Kearney & Porter, 2020):

Energy balance:

$$\dot{Q}_{solar} + \dot{Q}_{IR,in} + \dot{Q}_{metab} = \dot{Q}_{IR,out} + \dot{Q}_{conv} + \dot{Q}_{cond} + \dot{Q}_{evap}$$

Body temperature is found by root-solving $\sum \dot{Q}_{in} = \sum \dot{Q}_{out}$.

Water balance:

$$\text{Intake} = \text{Cutaneous} + \text{Respiratory} + \text{Excretion}$$

Coupled through evaporation and metabolism.

NicheMapR model features

- Driven by microclimate data (two shade extremes)
- Key functional traits: body size, shape, solar absorptivity, skin wetness, thermal tolerances
- Behavioural options: colour change, posture change, shade-seeking, climbing, panting, burrowing

Behavioural thermoregulation

The model implements a **deterministic state machine** with a fixed hierarchy of thermoregulatory responses:

- 1 **Emerge** — if $T_b > T_{RB,min}$ (minimum emergence temperature)
- 2 **Bask** — maximise silhouette area perpendicular to sun
- 3 **Forage** — if $T_{F,min} < T_b < T_{F,max}$ at T_{pref}
- 4 **Reduce absorptivity** — decrease α in 5% steps
- 5 **Change posture** — align parallel to sun's rays

Behavioural thermoregulation

- 1 **Seek shade** — up to maxshades
- 2 **Climb** — move to reference height (milder conditions)
- 3 **Pant** — increase respiratory airflow, oral evaporation
- 4 **Retreat underground** — find shallowest tolerable depth (SELDEP: between CT_{min} and $(T_{F,max} + CT_{max})/2$)

Key principle

Behaviour follows a **discrete sequence** of increasingly costly responses, triggered by body temperature thresholds — simple but ecologically realistic.

Steady-state vs transient

Aspect	NicheMapR (steady-state)	Rubalcaba (transient)
Heat balance	$\sum Q = 0$ each hour	dT_b/dt solved over time
Thermal inertia	Neglected	Explicit ($c_p M$)
Behaviour	Deterministic state machine	Probabilistic (multinomial logit)
Computation	Fast (root-finding)	Slower (ODE integration)
Best suited for	Small organisms	Large organisms

Both use the same physical principles, but differ in their treatment of time, behaviour, and computational cost.

From Tb to activity and development

Biophysical body temperature estimates feed into higher-level processes:

Activity windows:

- Three states: **inactive**, **basking**, **foraging**
- Foraging only between $T_{F,min}$ and $T_{F,max}$
- Activity hours per day $\rightarrow$ foraging opportunity
- Constrains energy intake and growth

Development:

- Degree-days above a developmental threshold
- Egg development driven by nest temperature
- Determines phenology and life-cycle completion
- Often the most thermally-sensitive life stage

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Key outputs

TC (body temperature), ACT (activity state: 0=inactive, 1=basking, 2=foraging), SHADE, DEP (burrow depth), enbal (energy balance), masbal (mass balance)

References I

- Gates, D. M. (1980). *Biophysical ecology* (D. E. Reichle, Red.). Springer New York.
<https://doi.org/10.1007/978-1-4612-6024-0>
- Helmuth, B. S. T. (1998). Intertidal mussel microclimates: Predicting the body temperature of a sessile invertebrate. *Ecological Monographs*, 68(1), 51–74.
[https://doi.org/10.1890/0012-9615\(1998\)068\[0051:IMMPTB\]2.0.CO;2](https://doi.org/10.1890/0012-9615(1998)068[0051:IMMPTB]2.0.CO;2)
- Kearney, M., & Porter, W. (2009). Mechanistic niche modelling: Combining physiological and spatial data to predict species' ranges. *Ecology Letters*, 12(4), 334–350.
<https://doi.org/10.1111/j.1461-0248.2008.01277.x>
- Kearney, M., & Porter, W. P. (2004). Mapping the fundamental niche: Physiology, climate, and the distribution of a nocturnal lizard. *Ecology*, 85(11), 3119–3131.
<https://doi.org/10.1890/03-0820>
- Kearney, M. R., & Porter, W. P. (2020). NicheMapR – an R package for biophysical modelling: The ectotherm and dynamic energy budget models. *Ecography*, 43(1), 85–96.
<https://doi.org/10.1111/ecog.04680>
- Porter, W. P., Mitchell, J. W., Beckman, W. A., & DeWitt, C. B. (1973). Behavioral implications of mechanistic ecology: Thermal and behavioral modeling of desert ectotherms and their microenvironment. *Oecologia*, 13(1), 1–54.
<https://doi.org/10.1007/BF00379617>
- Porter, W. P., & Gates, D. M. (1969). Thermodynamic equilibria of animals with environment. *Ecological Monographs*, 39(3), 227–244. <https://doi.org/10.2307/1948545>

References II

- Rubalcaba, J. G., Gouveia, S. F., & Olalla-Tárraga, M. A. (2019). A mechanistic model to scale up biophysical processes into geographical size gradients in ectotherms (A. Algar, Ed.). *Global Ecology and Biogeography*, 28(6), 793–803. <https://doi.org/10.1111/geb.12893>
- Seebacher, F., & Alford, R. A. (1999). Movement and microhabitat use of a terrestrial amphibian (*bufo marinus*) on a tropical island: Seasonal variation and environmental correlates. *Journal of Herpetology*, 33(2), 208. <https://doi.org/10.2307/1565716>
- Spotila, J. R., Lommen, P. W., Bakken, G. S., & Gates, D. M. (1973). A mathematical model for body temperatures of large reptiles: Implications for dinosaur ecology. *The American Naturalist*, 107(955), 391–404. <https://doi.org/10.1086/282842>
- Stevenson, R. D., Peterson, C. R., & Tsuji, J. S. (1985). The thermal dependence of locomotion, tongue flicking, digestion, and oxygen consumption in the wandering garter snake. *Physiological Zoology*, 58(1), 46–57. <https://doi.org/10.1086/physzool.58.1.30161219>
- Tracy, C. R. (1976). A model of the dynamic exchanges of water and energy between a terrestrial amphibian and its environment. *Ecological Monographs*, 46(3), 293–326. <https://doi.org/10.2307/1942256>